Matthew V. Pugh

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Detail-oriented software engineer with 4+ years of experience building automation tools through artificial intelligence and machine learning. Strong analytical skills and problem-solving abilities enable tackling complex technical challenges. Proven ability to adapt quickly and deliver measurable results across diverse team-based environments in government, consulting, and research sectors. Eager to leverage interdisciplinary expertise and passion for innovation to contribute to cutting-edge projects and grow professionally in a dynamic technical organization.

Work Experience

Cognizant IOT Software Engineer (SWE), May 2025 – Current:

- Launched AI/automation tools for MES/PASX XML data analysis with Cognizant APEx to streamline Generic Master Batch Record (GMBR) verification and instantiation. Tools reduced MBR validation time from weeks of manual work to minutes. Presented work to industry leaders across Cognizant.
- Developing integration and comparison of MBR environments across sites through AI analysis for compliance and efficient application of system setup; database querying and management through SQL across sites.
- Created JBoss log file analysis tool to support server administrators. Machine-learning-based extractive, analysis, summarization tool. Presented research into future agentic AI utilization for server uptime and management as tool extrapolation.

Federal Communication Commission (FCC) OET Intern, May 2024 – August 2024:

- Created dashboard to spatially and temporally model national Broadband Data Collection employed internally for FCC employees across the organization. R and python integration for data representation, analysis, and UX.
- Developed AI-based Large Language Model segmentation, extraction, and summarization tool for FCC Public Notices comment summarization. Conducted exploratory model analysis and selection, hyperparameter testing and tuning. Presented to FCC division leaders.
- Developed Machine Learning program to improve RF propagation modeling. Program will be integrated into PLUM (Propagation Modeling FCC internal software) and featured in an upcoming research paper.

Naval Information Warfare Center (NIWC) Pacific NREIP Intern, May 2023 – August 2023:

- Developed Generative Adversarial Network for use in generating synthetic EEG data for the purpose of establishing a brain computer interface. Presented research to various governmental departments including heads of communications, networks, and cybersecurity.

Skills

Programming Experience: Highly proficient industry-based experience with use in building automation tools, REST APIs, data analytics, machine learning, artificial intelligence, generative models through Python, Java, C, C++, R. Programming experience binary search trees, recursion, interfaces, exception handling, algorithm efficiency. Expertise with data structures including stacks, linked lists, graphs. Familiarity in app development, Android Studio.
Artificial Intelligence: Industry experience creating GAN, LSTM/GRU/RNN, Classification modeling, generative modeling, data analytics/deep learning. Experience with advanced models, OCR and computer vision tasks.
Computer Systems Security and Cybersecurity: Proficiency with privacy and security across systems, creating computer networks, securing systems, assessing systems for vulnerabilities. Published research in automating security operations through artificial intelligence.
SQL: Industry-based proficiency in design, optimization, querying of SQL databases and database management with Oracle Database, PL/SQL, Firebase.

Education

Johns Hopkins University Graduated May 2026
Master of Science in Security Informatics: Computer Programming, Data Analytics, Artificial Intelligence, Cybersecurity, Software Vulnerability Analysis, Critical Infrastructure Protection, Cryptography, Security and Privacy in Computing, Compliance and Regulation

Johns Hopkins University Graduated May 2025
Bachelor of Science in Computer Science: Human Language Theory, Computer Vision, User Interfaces and Mobile Applications, Algorithms, Computer System Fundamentals, Numerical Linear Algebra.
Bachelor of Science in Chemical Engineering: Dynamic Modeling and Processes, Kinetics, Process Safety, Thermodynamics, Transport Phenomena, Organic Chemistry, Lab Experience.

Honors/ Awards

Publication: AutoSOC Cyber Analyst (ASOC-CA): Using AI to Automate SOC Tier 1 & 2 Activities - Published in: 2026 IEEE 16th Annual Computing and Communication Workshop and Conference (CCWC). Authors: **Matthew Pugh**; Iason Mihalopoulos; Jen-Ryann Ngo-Antonio; Damon Neal; Yinzhi Cao; Lanier Watkins.
Johns Hopkins University Hodson Scholar, Fall 2021 – Spring 2026
Johns Hopkins University Dean's List, Fall 2021 – Spring 2026
Eagle Scout, February 2019